

Forecasting Bitcoin Prices and Analyzing Geopolitical Risks: A Residual Analysis Approach using LSTM

Technical Report 02 (TR-02):

Data Engineering Pipeline & Dataset Reconstruction

1. Abstract

This technical report documents the **Data Engineering** phase executed to prepare the Bitstamp dataset for the Long Short-Term Memory (LSTM) modeling phase. The primary objective was to transform raw, high-frequency transactional data into a consistent time-series dataset suitable for the project's target analysis period (**2012–2016**). The engineering pipeline addressed critical structural issues, including temporal formatting (Unix Timestamp conversion), unit scale anomalies (USD/BTC normalization), and high-frequency noise reduction via **Daily Resampling**. The final output is a clean, structured dataset strictly processed to serve as the input for the subsequent Deep Learning regression tasks.

2. Introduction

Following the conclusions of TR-01, which identified the limitations of traditional classifiers (Random Forest) in capturing temporal dependencies, the project has transitioned to a Deep Learning regression approach using LSTMs. This architecture is highly sensitive to input quality ("Garbage In, Garbage Out"). The raw Bitstamp dataset presented three major engineering challenges that necessitated a rigorous pre-processing pipeline:

1. **High-Frequency Noise:** The raw data contained minute-by-minute fluctuations. This level of granularity introduces stochastic noise that obscures the macroeconomic trends required for the project's geopolitical inference goals.
2. **Temporal Formatting:** The use of raw Unix Timestamps and non-standard CSV delimiters required a robust parsing strategy to ensure temporal alignment.
3. **Scale Anomalies:** A diagnostic check revealed that raw values were stored in cents (Fiat) and Satoshis (Crypto), magnitudes that would cause gradient instability (exploding gradients) during Neural Network training.

This report outlines the methodology used to curate the data specifically for the 2012–2016 analysis window.

3. Methodology

3.1. Data Ingestion & Temporal Reconstruction

The ingestion process involved correcting the semicolon (;) delimiter and reconstructing the timeline using **Unix Timestamps** rather than the string-based date column.

- *Technique:* We utilized `pd.to_datetime(unit='s')` to convert raw seconds into a precise DateTime index. This mathematical conversion eliminated ambiguity related to locale formats (e.g., DD/MM vs. MM/DD), ensuring zero temporal misalignment during the volatile periods of 2013 and 2014.

3.2. Automated Scale Correction & Numerical Stability

To ensure the LSTM architecture receives semantically correct and numerically stable inputs, we implemented a conversion logic based on the fundamental definitions of the currency units. This step

is critical because raw integer inputs with high magnitudes (e.g., volume in billions) can lead to **Gradient Explosion** during training.

3.2.1. Fiat Price Reconstruction (Cents to USD)

The diagnostic logic identified that raw prices (P_{raw}) were stored as integers representing **cents** to preserve floating-point precision in the source database.

- **Logic:** Since **1 USD = 100 Cents**, the conversion factor is naturally $\delta_{\text{fiat}} = 100$.
- **Application:** We applied the division to restore the standard USD value:

$$P_{\text{usd}} = \frac{P_{\text{raw}}}{100}$$

3.2.2. Crypto Volume Reconstruction (Satoshis to BTC)

Similarly, the trading volume (V_{raw}) was recorded in **Satoshis**, the smallest divisible unit of Bitcoin.

- **Logic:** By protocol definition, **1 Bitcoin = 100,000,000 (10^8) Satoshis**. The raw data presented magnitudes up to 10^{11} , which introduces severe variance.
- **Application:** To normalize the volume into standard BTC units V_{btc} , we applied the inverse power-of-ten scaling:

$$V_{\text{btc}} = \frac{V_{\text{raw}}}{10^8}$$

3.3. Granularity Reduction (Downsampling)

To optimize the model's ability to learn long-term dependencies without overfitting to intraday noise, we applied a **Daily Resampling** strategy.

- *Function:* `resample('D').agg(...)`
- *Aggregation Logic:*
 - **Open:** First transaction of the day.
 - **High:** Maximum price of the day.
 - **Low:** Minimum price of the day.
 - **Close:** Last transaction of the day (Target Variable).
 - **Volume:** Sum of all transactions.

4. Results & Data Discussion

4.1. Dataset Integrity (2012–2016 Scope)

The processing pipeline successfully reconstructed the complete daily history for the project's scope. The timeline is continuous, covering the early adoption phase (2012), the major 2013 bubble, and the subsequent correction in 2014, providing a rich variance for model training.

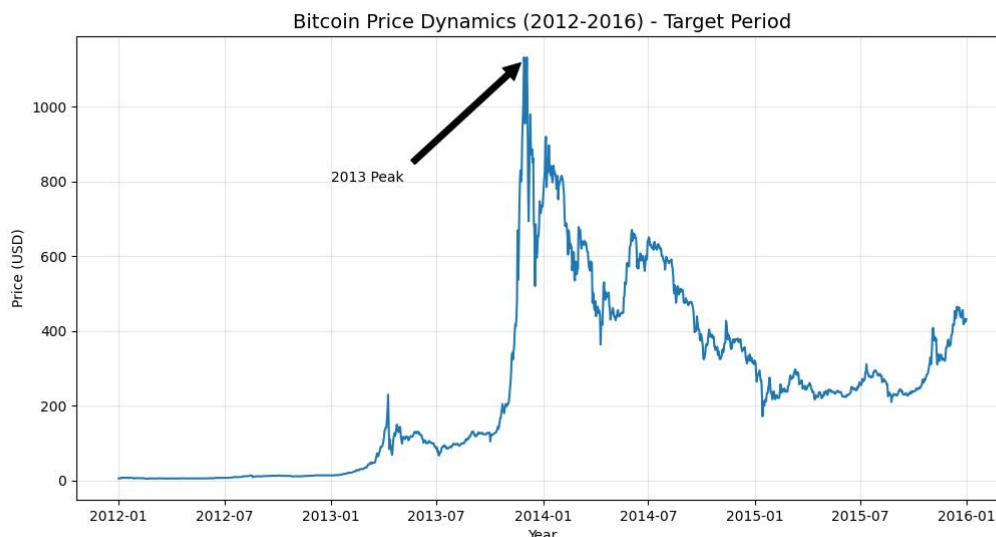


Figure 1: Reconstructed Bitcoin Price Dynamics (2012-2016). The dataset clearly captures the 2013 peak (~\$1,100) and the volatility of 2014, providing the necessary signal for the LSTM to learn market cycles.

4.2. Compression & Efficiency

The downsampling process reduced the dataset size by approximately **99.9%** (compressing millions of minute-rows into ~1,460 daily records). This drastic reduction removes high-frequency noise while preserving the signal intensity, ensuring the LSTM can be trained efficiently on standard GPU hardware.

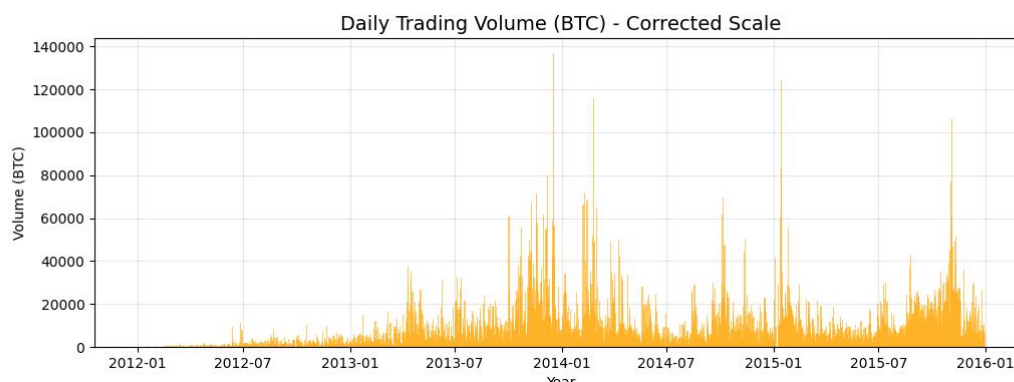


Figure 2: Daily Trading Volume (BTC). The volume scale is now consistent with real market liquidity, ready for feature engineering.

5. Conclusion

The Data Engineering phase has successfully concluded with the generation of the **bitstamp_daily_final.csv** dataset. By correcting the structural anomalies, restoring the correct unit scales (USD and BTC), and reducing the granular noise, we have established a robust foundation for the project.

This processed dataset is now fully compatible with the requirements of Phase 1, serving as the trusted input source for training the **LSTM Neural Network**. It ensures that the model will learn from accurate historical patterns to perform precise regression analysis on the validation period.